

# Relational Databases (contd.)

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Database System Concepts, 6th Ed.

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# Integrity Constraints

- Integrity constraints guard against accidental damage to the database, by ensuring that authorized changes to the database do not result in a loss of data consistency.
  - A checking account must have a balance greater than $10,000.00
  - A salary of a bank employee must be at least $4.00 an hour
  - A customer must have a (non-null) phone number

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# Integrity Constraints on a Single Relation

- **not null**
- **primary key**
- **unique**
- **check** (P), where P is a predicate

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# Not Null and Unique Constraints

- **not null**
  - Declare *name* and *budget* to be **not null**

    *name* **varchar**(20) **not null**
    *budget* **numeric**(12,2) **not null**
- **unique** ( $A_1$, $A_2$, …, $A_m$)
  - The unique specification states that the attributes

    $A1$, $A2$, … $Am$
    form a candidate key.
  - Candidate keys are permitted to be null (in contrast to primary keys).

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# The check clause

- **check** (P)

  where P is a predicate

  Example: ensure that semester value is one of fall, winter, spring or summer:

```
create table section (
    course_id varchar (8),
    sec_id varchar (8),
    semester varchar (6),
    year numeric (4,0),
    building varchar (15),
    room_number varchar (7),
    time slot id varchar (4),
    primary key (course_id, sec_id, semester, year),
    check (semester in ('Fall', 'Winter', 'Spring', 'Summer'))
);
```

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# Referential Integrity

- Ensures that a value that appears in one relation for a given set of attributes also appears for a certain set of attributes in another relation.
  - Example: If "Biology" is a department name appearing in one of the tuples in the *instructor* relation, then there exists a tuple in the *department* relation for "Biology".
- Let A be a set of attributes. Let R and S be two relations that contain attributes A and where A is the primary key of S. A is said to be a **foreign key** of R if for any values of A appearing in R these values also appear in S.

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# Cascading Actions in Referential Integrity

- **create table** *course (*
  *course_id* **char**(5),
  *title* **varchar**(20),
  *dept_name* **varchar**(20),
  **primary key** (*course_id*)
  **foreign key** (*dept_name*) **references** *department)*
- **create table** *course* (
  …
  *dept_name* **varchar**(20),
  **foreign key** (*dept_name*) **references** *department*
  **on delete cascade**
  **on update cascade**,
  . . .
  )
- alternative actions to cascade: **set null**, **set default**

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# Triggers

- A **trigger** is a statement that is executed automatically by the system as a side effect of a modification to the database.
- To design a trigger mechanism, we must:
  - Specify the conditions under which the trigger is to be executed.
  - Specify the actions to be taken when the trigger executes.
- Triggers introduced to SQL standard in SQL:1999, but supported even earlier using non-standard syntax by most databases.
  - Syntax illustrated here may not work exactly on your database system; check the system manuals

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# Trigger Example

- E.g. *time_slot_id* is not a primary key of *timeslot,* so we cannot create a foreign key constraint from *section* to *timeslot.*
- Alternative: use triggers on *section* and *timeslot* to enforce integrity constraints

```
create trigger timeslot_check1 after insert on section
referencing new row as nrow
for each row
when (nrow.time_slot_id not in (
            select time_slot_id
            from time_slot)) /* time_slot_id not present in time_slot */
begin
    rollback
end;
```

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# Trigger Example Cont.

```
create trigger timeslot_check2 after delete on timeslot
    referencing old row as orow
    for each row
    when (orow.time_slot_id not in (
                select time_slot_id
                from time_slot)
                /* last tuple for time slot id deleted from time slot */
            and orow.time_slot_id in (
                select time_slot_id
                from section)) /* and time_slot_id still referenced from section*/
    begin
       rollback
    end;
```

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# Triggering Events and Actions in SQL

- Triggering event can be **insert**, **delete** or **update**
- Triggers on update can be restricted to specific attributes
  - **E.g., after update of** *takes* **on** *grade*
- Values of attributes before and after an update can be referenced
  - **referencing old row as** **:** for deletes and updates
  - **referencing new row as** **:** for inserts and updates
- Triggers can be activated before an event, which can serve as extra constraints. E.g. convert blank grades to null.

```
create trigger setnull_trigger before update of takes
referencing new row as nrow
for each row
when (nrow.grade = ' ')
begin atomic
        set nrow.grade = null;
end;
```

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# Trigger to Maintain credits_earned value

- **create trigger** *credits_earned* **after update of** *takes* **on** (*grade*)
  **referencing new row as** *nrow*
  **referencing old row as** *orow*
  **for each row**
  **when** *nrow.grade* <> 'F' **and** *nrow.grade* **is not null**
  **and** (*orow.grade* = 'F' **or** *orow.grade* **is null**)
  **begin atomic**
  **update** *student*
  **set** *tot_cred*= *tot_cred* +
  (**select** *credits*
  **from** *course*
  **where** *course.course_id*= *nrow.course_id*)
  **where** *student.id* = *nrow.id*;
  **end**;

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# When Not To Use Triggers

- Triggers were used earlier for tasks such as
  - maintaining summary data (e.g., total salary of each department)
  - Replicating databases by recording changes to special relations (called **change** or **delta** relations) and having a separate process that applies the changes over to a replica
- There are better ways of doing these now:
  - Databases today provide built in materialized view facilities to maintain summary data
  - Databases provide built-in support for replication
- Encapsulation facilities can be used instead of triggers in many cases
  - Define methods to update fields
  - Carry out actions as part of the update methods instead of through a trigger
- Risk of unintended execution of triggers, for example, when
  - loading data from a backup copy
  - replicating updates at a remote site
  - Trigger execution can be disabled before such actions.
- Other risks with triggers:
  - Error leading to failure of critical transactions that set off the trigger
  - Cascading execution



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# Views

- In some cases, it is not desirable for all users to see the entire logical model (that is, all the actual relations stored in the database.)
- Consider a person who needs to know an instructors name and department, but not the salary. This person should see a relation described, in SQL, by

  **select** *ID*, *name*, *dept_name*
  **from** *instructor*

- A **view** provides a mechanism to hide certain data from the view of certain users.
- Any relation that is not of the conceptual model but is made visible to a user as a "virtual relation" is called a **view**.

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# View Definition

- A view is defined using the **create view** statement which has the form

  **create view** *v* **as** < query expression >

  where <query expression> is any legal SQL expression. The view name is represented by *v.*
- Once a view is defined, the view name can be used to refer to the virtual relation that the view generates.
- View definition is not the same as creating a new relation by evaluating the query expression
  - Rather, a view definition causes the saving of an expression; the expression is substituted into queries using the view.


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# Example Views

- A view of instructors without their salary

  **create view** *faculty* **as**
  **select** *ID*, *name*, *dept_name*
  **from** *instructor*

- Find all instructors in the Biology department

  **select** *name*
  **from** *faculty*
  **where** *dept_name* = 'Biology'

- Create a view of department salary totals

  **create view** *departments_total_salary*(*dept_name*, *total_salary*) **as**
  **select** *dept_name*, **sum** (*salary*)
  **from** *instructor*
  **group by** *dept_name*;

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# Views Defined Using Other Views

- **create view** *physics_fall_2009* **as**
  **select** *course.course_id*, *sec_id*, *building*, *room_number*
  **from** *course*, *section*
  **where** *course.course_id* = *section.course_id*
  **and** *course.dept_name* = 'Physics'
  **and** *section.semester* = 'Fall'
  **and** *section.year* = '2009';
- **create view** *physics_fall_2009_watson* **as**
  **select** *course_id*, *room_number*
  **from** *physics_fall_2009*
  **where** *building*= 'Watson';

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# View Expansion

- Expand use of a view in a query/another view

```
create view physics_fall_2009_watson as
(select course_id, room_number
from (select course.course_id, building, room_number
        from course, section
        where course.course_id = section.course_id
            and course.dept_name = 'Physics'
            and section.semester = 'Fall'
            and section.year = '2009')
where building= 'Watson';
```

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# Update of a View

- Add a new tuple to *faculty* view which we defined earlier

  **insert into** *faculty* **values** (’30765’, ’Green’, ’Music’);

  This insertion must be represented by the insertion of the tuple

  (’30765’, ’Green’, ’Music’, null)

  into the *instructor* relation

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# Some Updates cannot be Translated Uniquely

- **create view** *instructor_info* **as**
  **select** *ID*, *name*, *building*
  **from** *instructor*, *department*
  **where** *instructor.dept_name= department.dept_name*;
- **insert into** *instructor_info* **values** ('69987', 'White', 'Taylor');
  - which department, if multiple departments in Taylor?
  - what if no department is in Taylor?
- Most SQL implementations allow updates only on simple views
  - The **from** clause has only one database relation.
  - The **select** clause contains only attribute names of the relation, and does not have any expressions, aggregates, or **distinct** specification.
  - Any attribute not listed in the **select** clause can be set to null
  - The query does not have a **group** by or **having** clause.



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# More Problems

- **create view** *history_instructors* **as**
  **select** *
  **from** *instructor*
  **where** *dept_name*= ’History’;
- What happens if we insert (’25566’, ’Brown’, ’Biology’, 100000) into *history_instructors?*

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# Materialized Views

- Certain database systems allow view relations to be stored
- View Maintenance
  - If the actual relations used in the view definition change, the view is kept up to date.
- Applications that use a view frequently benefit from materialized views
- Fast response for view-based queries

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